AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Patient Prioritization for Faster, Fairer Hospital Care

Team ID: *GEN26-AOH-I097*

Team Name: *THE DISRUPTORS*

Domain: *Domain 6: Healthcare, Biotechnology & Human Well-being*

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Presentation Outline

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Problem Statement

Patient Prioritization: an intelligent system that reduces hospital waiting times by prioritizing patients according to urgency, symptoms, and available resources.

Problem Understanding

- Patients and attendants in OPD and emergency departments experience long, uncertain waits.
- A token number indicates arrival order, not the medical urgency of the person waiting.
- Triage is often manual, while doctor availability, department load, and bed status may be recorded separately.
- Patients may not know their expected waiting time or why another patient is called first.
- Critical cases can be delayed, waiting areas become crowded, and staff must repeatedly manage queue conflicts.

Existing Gaps & Challenges

- Manual triage can be slow and may vary between staff members during busy periods.
- Basic token systems cannot automatically react to worsening symptoms or changing resource availability.
- Patient, doctor, department, and bed information may sit in separate records, making coordination difficult.
- Waiting-time estimates are often missing or not updated when the queue changes.
- The gap: one live view that combines urgency, symptoms, available resources, and clear queue status.

Our Proposed Solution

- Register each patient with key symptoms and relevant intake details; optional vital signs can support assessment.
- Classify cases as **Critical, Urgent, or Non-urgent** using clinician-validated triage rules.
- Reorder the queue when urgency or doctor, department, and bed availability changes.
- Display priority, estimated wait, and resource status to authorised staff.
- Nurses and doctors can review and override priorities; the system supports, not replaces, clinical judgement.



Figure: Live priority queue dashboard

Technical Approach

Patient Intake → Triage Scoring Engine → Priority Queue → Staff Dashboard.

Inputs: symptoms, optional vital signs, doctors, departments, and beds.

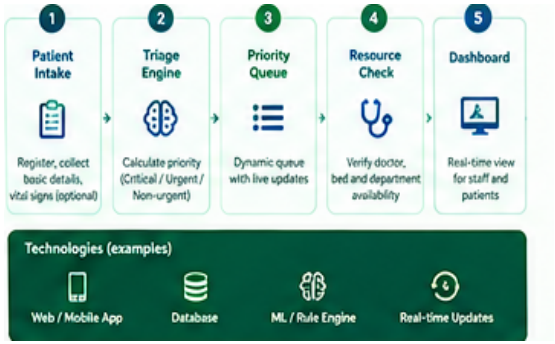


Figure: workflow

Staff enter patient details; a rule-based triage engine assigns urgency, then combines it with current resource availability to update queue order and estimated waiting time. Nurses and doctors validate urgency decisions.

Tech Stack

Frontend: Web app / staff dashboard

Backend: Triage scoring engine

Database: Patient, doctor, bed, department records

AI/ML, APIs, or Hardware: Rule-based first; ML-ready; optional BP, SpO2, temperature devices

Innovation & Uniqueness

- **Urgency-first, not token-first:** prioritises clinical need instead of arrival order alone.
- **Resource-aware:** considers doctor availability, department load, and bed availability together.
- **Dynamic updates:** allows the queue to change when symptoms or hospital capacity changes.
- **Transparent:** provides staff with a shared view and gives patients clearer waiting-time expectations.
- **Human-supervised:** triage rules are validated by clinicians; the tool assists rather than replaces medical judgement.

Feasibility & Scalability

- **Minimum viable prototype:** front-desk PC/tablet, a basic database, rule-based scoring, and a staff dashboard.
- **People needed:** developers and UI designer, plus nurses/doctors to validate urgency rules and sample scenarios.
- **Scale-up path:** begin with one department, then add departments and connect doctor, bed, and workload data.
- **Key risks:** incomplete intake data, outdated resource status, and inappropriate triage rules.
- **Mitigation:** clinician review, anonymised test data, staff override, audit logs, and regular rule checks.

Cost for Product (Optional)

- Development or hardware cost: To be estimated after prototype scope and hardware choices are finalised.
- Hosting or maintenance cost: To be estimated; depends on deployment and maintenance requirements.
- Estimated total: Not specified in the source material.

Use Cases (compulsorily 3 use cases must be there)

- Use case 1 — Emergency intake: a patient assessed as Critical is escalated ahead of routine token numbers for immediate staff attention.
- Use case 2 — OPD queue: patients classified as Urgent are prioritised over Non-urgent cases, with updated estimated waiting times.
- Use case 3 — Resource-aware routing: staff view doctor availability, department load, and bed availability when managing the queue.

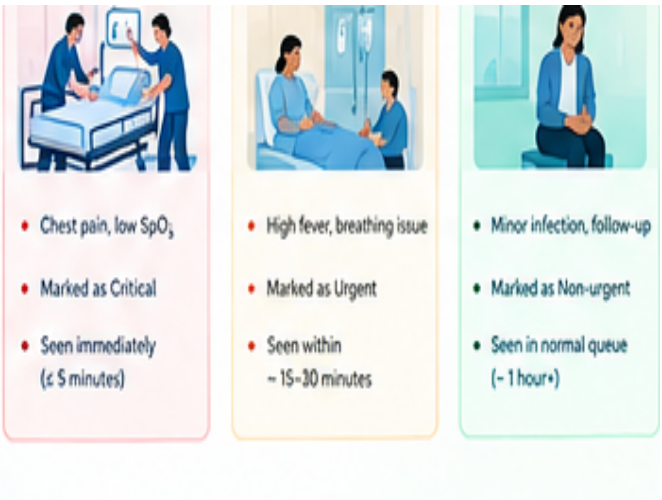


Figure: uses

Example walkthrough

1. If a waiting patient becomes more unwell, staff reassess symptoms and the case is flagged for immediate clinical review.
2. If a department is overloaded, the dashboard shows queue and department load to support coordination.

Impacts and Benefits

- **Patient safety:** faster escalation for critical cases can reduce avoidable delays in assessment.
- **Patient experience:** clearer queue status and estimated waits reduce uncertainty for patients and attendants.
- **Staff efficiency:** a shared dashboard can reduce manual queue coordination and repeated status checks.
- **Resource use:** better visibility of doctors, departments, and beds supports more coordinated decisions.
- **Demo metrics:** critical/urgent waiting time, overall waiting time, queue update speed, and resource utilisation.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 3	Target 3.8	Good Health and Well-being	Supports timely access to essential healthcare services.
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	Applies digital systems to improve healthcare service infrastructure.

Individual Member Contribution

Member	Contribution or responsibility
Adityaa Vaishnave K	To be confirmed by the team.
Anish U K	To be confirmed by the team.
Mohnish Kumar R B	To be confirmed by the team.
Aathithya S R	To be confirmed by the team.

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